Title	Predicting trial to paid user conversion on video streaming platforms by using session based behavioural data
Harvard referencing	(Kostovska and Gjoreski, 2025)
Link	https://icaiit.org/proceedings/13th_ICAIIT_5/3-3-ICAIIT_2025_13(5).pdf
Modelling	Logistic Regression Decision Trees Knearest neighbours(Knn) Random Forest Gradient Boosting Ensemble modes (the Random Forest and Gradient boosting)has the best F1 score and recall for the minority class which was converted
Preprocessing techniques	They removed sensitive attributes such as regional info, the unique identifiers . They also excluded users who were recorded not having session activity. The aggregated session level logs on user level features such as total sessions, the duration (total, average, max), unique devices and unique channels. The were label encoded categorical fields(OS and device type) They handles class imbalance through SMOTE, random under sampling(to 10 000 majority samples)
Data used	There is propriety behavioural data used from a global streaming platform that has 60 000 trial users which is 10% of users. The data has metadata, session activity logs and conversion outcomes that are not publicly available.
Key takeaways	The papers shows that tree based ensembles are the way to go for conversion scenarios. The Random Forst and Gradient boosting outperformed simpler models. This would mean it would be ideal for me to use Random Forest as my main model while keeping logistic Regression as a baseline. Their feature engineering approach by turning raw session logs into toal sessions, watch time and number of devices is the approach I would want to take for StadioChoice event data instead of just basing it on age, gender and such variables. The 10% converters was show an imbalanced data set and what worked was SMOTE and undersampling to handle it, which is a big consideration is my conversion rate is low too.

Title	A data driven approach to predict the success of bank telemarketing
Harvard referencing	(Moro et al., 2014)
Link	https://www.sciencedirect.com/science/article/abs/pii/S016792361400061X
Modelling	Logistic Regression Decision Tree Neural Network Spport Vector Machine
Preprocessing techniques	The technique used was starting from 150 raw features and then recuded to 22 through semi automatic feature selection. They used a rolling window evaluation scheme and this was trained on data before 2012 and tested on data after so that is mimics a more realistic future prediction instead of random splitting
Data used	The data used was Bank Marketing dataset from a Portuguese retail bank between the years 2008 and 2013 and its publicly available on UCI Machine learning repository
Key takeaways	They went from 150 raw features to 22 which shows not to just dump every column I have into the model, I must have a good feature selection and focus on the variables that align well with conversion. A consideration for me is how they had data splitting by splitting my data by sign up date instead of having a random split. They also compared different models rather than just sticking to one, which is a big consideration for my plans to compare Logistic Regression and Random Forest instead of just picking one as the better one.

Title	A comparison of resampling techniques to handle the class imbalance problem in machine learning : Conversion Prediction of Spotify users
Harvard referencing	(Jagelid and Movin, 2017)
Link	https://kth.diva-portal.org/smash/get/diva2:1108465/FULLTEXT02.pdf
Modelling	Logistic Regression Decision Tree Gradient Boosting The Logistic Regression and Gradient Boosting with Random oversampling (50-50) had the best AUC performance
Preprocessing techniques	The categorical features are one hot encoded. Normalised all features by using training set mean and variance. The class imbalance was treated by Random Oversampling (ROS), Random under sampling and SMOTE at 30/70 and 50/50 ratio, the data split into train 2-3 and test 1-3 per daily cohort
Data used	It is private Spotify user data that has 523 838 free users that were registered over 10 days in early 2017, 2% of those converted to premium within 14 days. The features have user activity such as sessions, and consumed hours , the demographics such as country and platform. The data us not publicly available
Key takeaways	Their data only has 2% converts which is a clear imbalanced data while the Logistic Regression and Gradient boosting with oversampling has the best AUC, which shows I should resample rather than not consider the imbalance. They had features like the number of sessions, total session length and the number of active days from raw logs which is the same thinking around my feature engineering I want to do for StadioChoice by the number of logins, total watch time and content diversity. They also has AUC instead of accuracy as the metric they rely on which makes sense for such an imbalance, which is something I should keep in mind as well as the F1and recall.

Dataset for partB

Name	Bank Marketing
Harvard referencing	(Moro et al., 2014)
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41 188, one dataset with no train test split
Number of features columns	21, includes the target column
Number of classes	2, (Whether subscribed: yes or no)
Target range	It's a classification problem but not transgression so it wont have.
Origin	From a direct marketing campaign data though a Portuguese retail bank compiled between the years 2008 to 2013 and released publicly through the UCI Machine learning repository

References:

JAGELID, T. & MOVIN, J. 2017. *From Free to Fee: Converting Freemium Users into Premium Subscribers*. Bachelor's thesis, Lund University.

KOSTOVSKA, A. & GJORESKI, F. 2025. Predicting Subscription Conversion from Free Trial Usage Data. *arXiv preprint*.

MORO, S., CORTEZ, P. & RITA, P. 2014. A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems,* 62**,** 22–31.